

AI Assisted Coding Assignment (Lab 15.3

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****Lab Assignment 15.3 – AI Assisted Coding****

****Task 1: Online Food Ordering API****

****Question****

Design a REST API for an online food ordering system with CRUD operations.

****Prompt (used with AI)****

"Generate a Python Flask REST API for an online food ordering system with endpoints for menu listing, placing orders, updating, tracking, and deleting orders. Include comments and basic validation."

****Code****

```
from flask import Flask, jsonify, request
```

```
app = Flask(__name__)
```

```
menu = [
```

```
    {"id": 1, "name": "Pizza", "price": 250},
```

```
    {"id": 2, "name": "Burger", "price": 120}
```

```
]
```

```
orders = []
```

```
@app.route('/menu', methods=['GET'])
```

```
def get_menu():
```

```
    return jsonify(menu)
```

```
@app.route('/order', methods=['POST'])
```

```
def place_order():
```

```
    data = request.json
```

```
    order = {"id": len(orders)+1, "items": data["items"], "status": "Placed"}
```

```
    orders.append(order)
```

```
    return jsonify(order)
```

```
@app.route('/order/<int:id>', methods=['GET'])
```

```
def get_order(id):
```

```
    for order in orders:
```

```
        if order["id"] == id:
```

```
            return jsonify(order)
```

```
    return jsonify({"error": "Order not found"}), 404
```

```
@app.route('/order/<int:id>', methods=['PUT'])
```

```
def update_order(id):
```

```
    data = request.json
```

```
    for order in orders:
```

```
        if order["id"] == id:
```

```
            order["items"] = data["items"]
```

```
            return jsonify(order)
```

```
    return jsonify({"error": "Order not found"}), 404
```

```
@app.route('/order/<int:id>', methods=['DELETE'])
def delete_order(id):
    for order in orders:
        if order["id"] == id:
            orders.remove(order)
            return jsonify({"message": "Order deleted"})
    return jsonify({"error": "Order not found"}), 404

app.run(debug=True)
```

Output

GET /menu →

```
[
  {"id":1,"name":"Pizza","price":250},
  {"id":2,"name":"Burger","price":120}
]
```

POST /order →

```
{"id":1,"items":["Pizza"],"status":"Placed"}
```

****Code Explanation****

- * Flask is used to create REST API.
- * `/menu` returns list of dishes.
- * `/order` creates a new order.
- * `/order/<id>` supports GET, PUT, DELETE.
- * Data stored in in-memory lists.

****Task 2: Employee Payroll API****

****Question****

Create an API to manage employees and salaries.

****Prompt****

"Generate a Flask API for employee payroll management with endpoints for adding employees, updating salary, deleting employees, and listing all employees."

****Code****

```
from flask import Flask, jsonify, request

app = Flask(__name__)

employees = []

@app.route('/employees', methods=['GET'])
def get_employees():
    return jsonify(employees)

@app.route('/employees', methods=['POST'])
def add_employee():
    data = request.json
    emp = {
        "id": len(employees)+1,
        "name": data["name"],
```

```

        "salary": data["salary"]
    }
    employees.append(emo)
    return jsonify(emp)

@app.route('/employees/<int:id>/salary', methods=['PUT'])
def update_salary(id):
    data = request.json
    for emo in employees:
        if emp["id"] == id:
            emp["salary"] = data["salary"]
            return jsonify(emp)
    return jsonify({"error": "Employee not found"}), 404

@app.route('/employees/<int:id>', methods=['DELETE'])
def delete_employee(id):
    for emo in employees:
        if emp["id"] == id:
            employees.remove(emp)
            return jsonify({"message": "Employee deleted"})
    return jsonify({"error": "Employee not found"}), 404

app.run(debug=True)

```

****Output****

POST /employees →

```
{"id":1,"name":"John","salary":50000}
```

PUT /employees/1/salary →

```
{"id":1,"name":"John","salary":60000}
```

****Code Explanation****

- * Stores employee data in list.
- * Supports CRUD operations.
- * Salary update handled separately.
- * Includes validation for missing employee.

****Task 3: Student Information REST API****

****Question****

Build a Student Management API.

****Prompt****

"Create a Flask REST API for student management with add, get, update, and delete operations."

****Code****

```
from flask import Flask, jsonify, request

app = Flask(__name__)

students = []

@app.route('/students', methods=['POST'])
def add_student():
    data = request.json
    student = {
        "id": len(students)+1,
        "name": data["name"],
```

```

        "course": data["course"]
    }
    students.append(student)
    return jsonify(student)

@app.route('/students', methods=['GET'])
def get_students():
    return jsonify(students)

@app.route('/students/<int:id>', methods=['PUT'])
def update_student(id):
    data = request.json
    for s in students:
        if s["id"] == id:
            s["name"] = data["name"]
            return jsonify(s)
    return jsonify({"error": "Not found"}), 404

@app.route('/students/<int:id>', methods=['DELETE'])
def delete_student(id):
    for s in students:
        if s["id"] == id:
            students.remove(s)
            return jsonify({"message": "Deleted"})
    return jsonify({"error": "Not found"}), 404

app.run(debug=True)

```

****Output****

POST /students →

```
{"id":1,"name":"Alice","course":"CSE"}
```

GET /students →

```
[{"id":1,"name":"Alice","course":"CSE"}]
```

****Code Explanation****

- * Manages student records.
- * Uses list as database.
- * CRUD operations implemented.
- * Each student has unique ID.

****Task 4: Weather Information API****

****Question****

Integrate external weather API.

****Prompt****

"Generate Flask API code to fetch current weather using OpenWeatherMap API."

****Code****

```
import requests
```

```
from flask import Flask, jsonify
```

```
app = Flask(__name__)
```

```
API_KEY = "sk-proj-
```

```
vMZZoXTmGybZEzckB0RBCrBffi374IYS1gXN9cBuc3ZVd649SBCoRzx_D0i8AcXIB4EOmTs3V6T3BlbkFJffu  
tlCXnL37pRSaAdmc6tWO9OHQWusNZCcnzOhsgaybRnLb9SdQiaY5HJNJUmjfiisnOk_T44A"
```



```
@app.route('/weather/current/<city>', methods=['GET'])

def get_weather(city):

    url = f"http://api.openweathermap.org/data/2.5/weather?q={city}&appid={sk-proj-
vMZZoXTmGybZEzckB0RBCrBffl374lYS1gXN9cBuc3ZVd649SBCoRzx_D0i8AcXlB4EOmTs3V6T3BlbkFJffu
tlCXnL37pRSaAdmc6tWO9OHQWusNZCcnzOhsgaybRnLb9SdQiaY5HJNJUmjfiisnOk_T44A}"

    response = requests.get(url).json()

    return jsonify({

        "temperature": response["main"]["temp"],

        "humidity": response["main"]["humidity"],

        "condition": response["weather"][0]["description"]

    })
```

****Output****

GET /weather/current/Chennai →

```
{
  "temperature": 300,
  "humidity": 70,
  "condition": "clear sky"
}
```

****Code Explanation****

- * Uses requests library to call API.
- * Extracts temperature, humidity, condition.
- * Returns structured JSON response.

****Task 5: Delete Item API****

****Question****

Create DELETE endpoint to remove item by index.

****Prompt****

"Write Flask DELETE API to remove item from list using index with validation."

****Code****

```
```python
from flask import Flask, jsonify

app = Flask(__name__)

items = [{"name": "Notebook", "price": 250}]

@app.route('/items/<int:index>', methods=['DELETE'])
def delete_item(index):
 if index < 0 or index >= len(items):
 return jsonify({"error": "Item not found"}), 404

 removed_item = items.pop(index)
 return jsonify({"message": "Item deleted", "item": removed_item})

@app.route('/items', methods=['GET'])
def get_items():
 return jsonify(items)

app.run(debug=True)
```

### **\*\*Output\*\***

DELETE /items/0 →

```
{"message": "Item deleted", "item": {"name": "Notebook", "price": 250}}
```

GET /items →

[]

**\*\*Code Explanation\*\***

- \* Validates index range.
- \* Removes item using pop().
- \* Returns deleted item.
- \* Ensures error handling.